

CUSTOMER MILESTONE BRIEF

MS #41 - HLASM Qualification Hardening

Purpose, customer value, delivered capability, evidence, and claim boundaries

Status	Complete
Release	0.41.0
Date	2026-08-31
Audience	Customers, partners, executives, architects, delivery teams, and assurance reviewers

Executive summary

MS #41 - HLASM Qualification Hardening converts a specific part of the LIGHTYEAR modernization approach into a governed, reviewable capability. Qualifies a bounded assembler subset and its COBOL handoff while explicitly excluding native assembler, binder, timing, and privileged-service claims.

Why this matters to a customer

Qualifies a bounded assembler subset and its COBOL handoff while explicitly excluding native assembler, binder, timing, and privileged-service claims.

The practical result is a narrower, evidence-backed decision instead of a broad modernization claim. Commercial, architecture, delivery, security, and assurance teams can review the same capability, proof, and limits.

What the milestone delivers

- Bound HLASM qualification to the canonical graph inventory of two programs, 41 instructions, 23 symbols, one DSECT, five fields, one macro, nine branch edges, and exact DSECT/macro relationships.
- Added a deterministic 40-case synthetic semantic corpus with 32 targeted boundary cases, 36 passing cases, four explicit fail-closed cases, and 121 observed feature categories.
- Covered COBDATFT date behavior and source quirks, the 80-byte COCDATFT layout, MVC/MVI/ST, CLI/CLC condition codes, B/BE/BNE, register operations, STM/LM save areas, USING/DROP, COPY, ASMWAIT static expansion, LTOrg, COBOL parameter lists, and the bounded MVSWAIT handoff.
- Added a 28-entry HLASM compatibility ledger using all five governing classifications and explicit exclusions for privileged services, storage protection, recovery, broad

instruction families, native assembler output, binder behavior, and STIMER timing.

- Separated eleven gates for graph inventory, corpus provenance, source/directives, DSECT/storage, instructions, condition codes/branches, linkage policy, native exclusions, the existing private date proof, and authorized native build/execution evidence.
- Added frozen schemas, POSIX/PowerShell release-gate integration, deterministic receipts, graph holdouts, exact-corpus checks, negative diagnostics, and rehashed-overclaim tests.
- Kept native HLASM, assembler, binder, LE linkage, system services, runtime, mainframe-equivalence, and production claims false pending authorized assembly, bind, execution, and operational evidence.

How it differs from earlier work

MS #41 builds on MS #40 (IMS Qualification Hardening). The earlier milestone established its own bounded capability; this milestone adds hlasm qualification hardening while preserving the earlier evidence and its limits.

Earlier evidence is not automatically promoted into this milestone. A parser, simulator, local comparison, or generated artifact is not live platform equivalence or production readiness.

What a customer can do with it

A customer can use hlasm qualification hardening as a bounded decision and delivery capability, attached to approved sources, owners, policies, target architecture, acceptance criteria, and authorized evidence.

Unresolved semantic loss, policy decisions, unsupported behavior, and live-evidence requirements remain visible.

Evidence and acceptance posture

The repository capability is complete because its committed implementation and release record are present. Customer qualification remains claim-specific and evidence-class-specific.

Review exact source and artifact identities, distinguish development from authorized live evidence, inspect policy and loss classifications, confirm passed and blocked gates, and verify that later changes have not invalidated the evidence.

Boundaries and claims not made

- Kept native HLASM, assembler, binder, LE linkage, system services, runtime, mainframe-equivalence, and production claims false pending authorized assembly, bind, execution, and operational evidence.

Source of record

- CHANGELOG.md release section(s): 0.41.0

- README.md product and operating guidance
- LIGHTYEAR-ROADMAP.md governing sequence and claim classifications
- docs/milestones/catalog.json metadata and docs/milestones/manifest.json artifact integrity

The underlying schemas, receipts, ledgers, tests, and policy decisions remain authoritative when greater detail is required.